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Linux Administration
2/1/26

Server Documentation

Prior to this assignment I had never written documentation. I saw the example video on blackboard and looked up some different ways to do it. For this documentation I have created, I wrote it the best I could with being as clear as possible.

Below are the two different distros I used for this server assignment. The first was **CentOS** and the second was **Ubuntu. I added the links to the iso's I chose.**

Question **1:**

I chose **two different** Linux Distros. I didn't know what **distros** were until researching more about it and speaking to a tutor about the correct setup and methods. **Below I write down the setup and also provide screenshots for the setup, along with showing what you can do within the virtual machines created. I tried my best to format the pictures and text together but it might be split up a little bit as there were errors with adding photos to the document and having text under the image.**

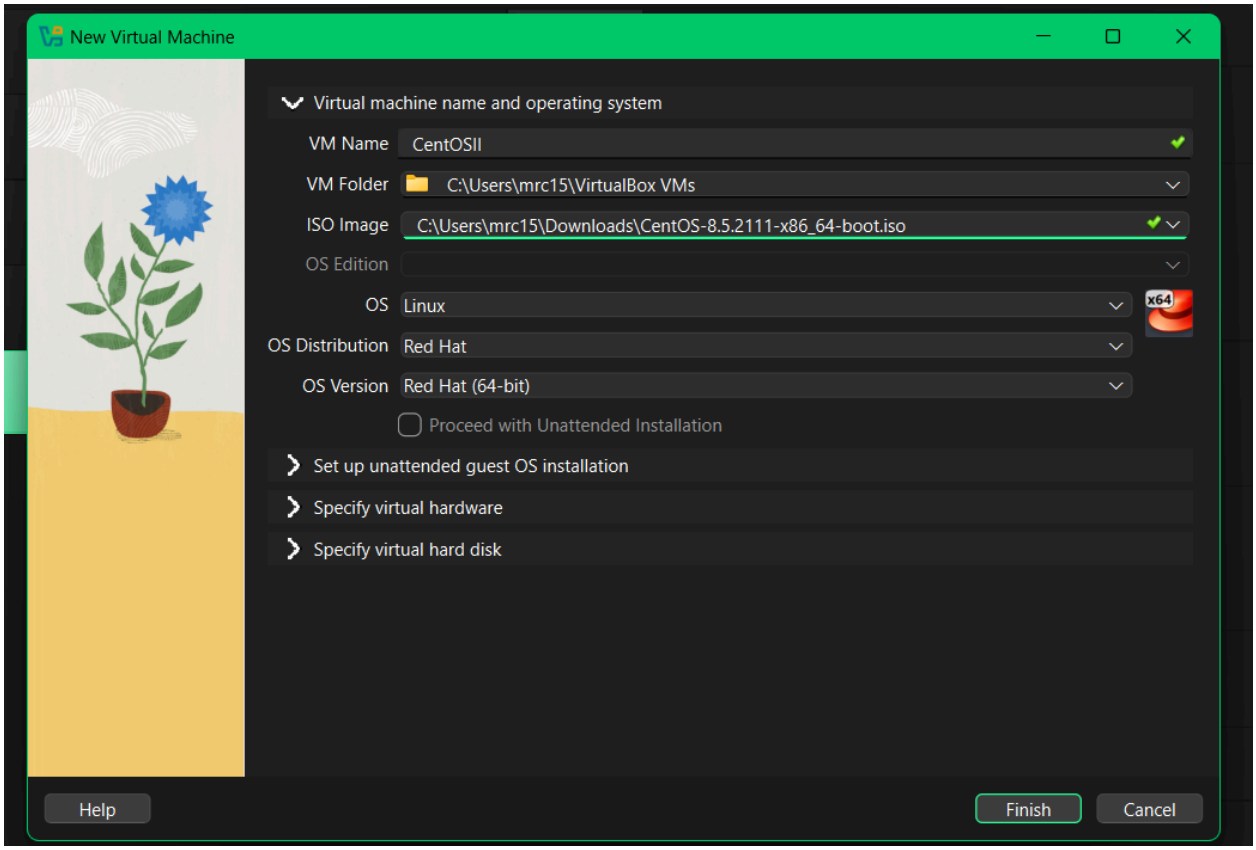
Question **2:**

CentOS Server Source- http://ftp.cs.stanford.edu/centos/centos/8.5.2111/isos/x86_64/

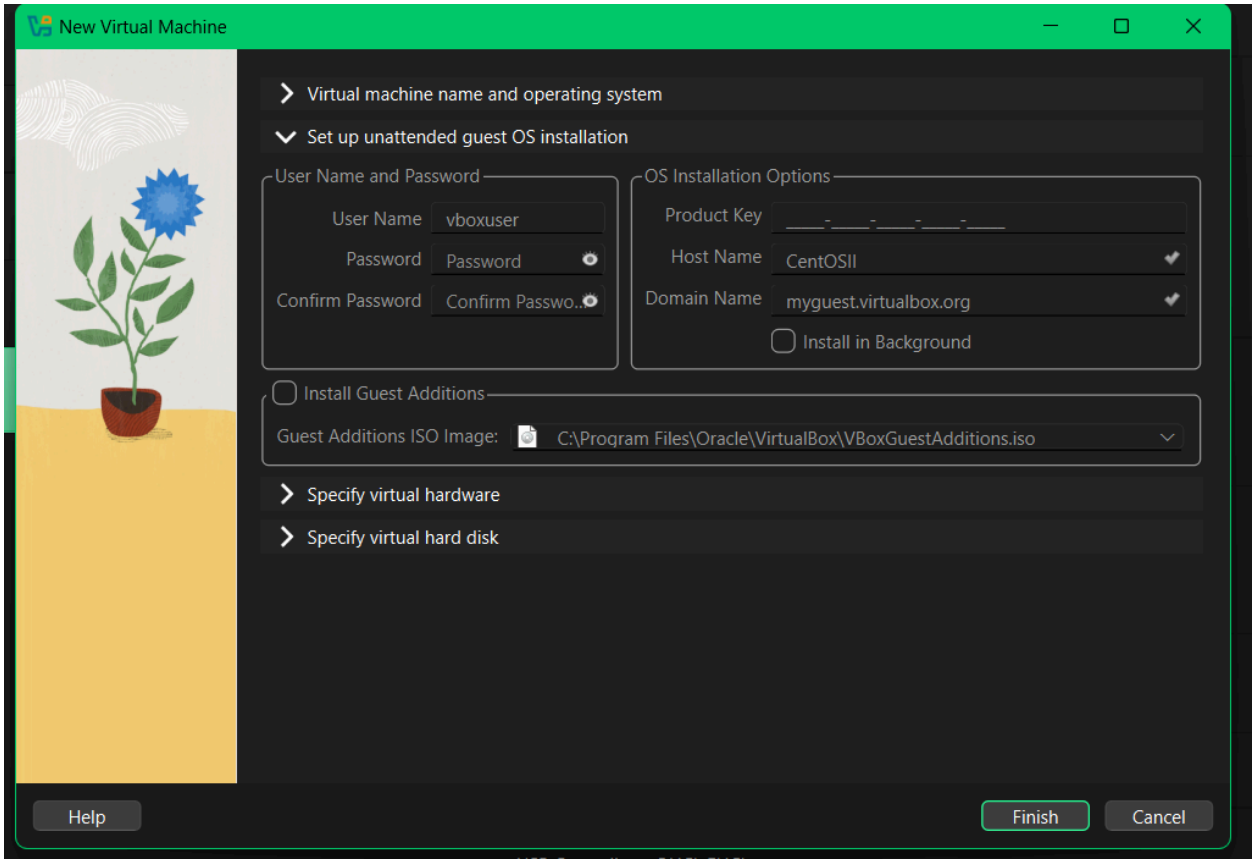
For the **CentOS** source it took me a while to get it to work. About two hours of trial and error. Adjusting different settings within the virtual box environment. The first issue I encountered was t**he discs size, the overall software space and speed of installation adjusting.** I continued to work through the errors until different settings worked properly.

Below I have added some screenshots of the setup I used for CentOS, with detailed instructions on how to create the server.

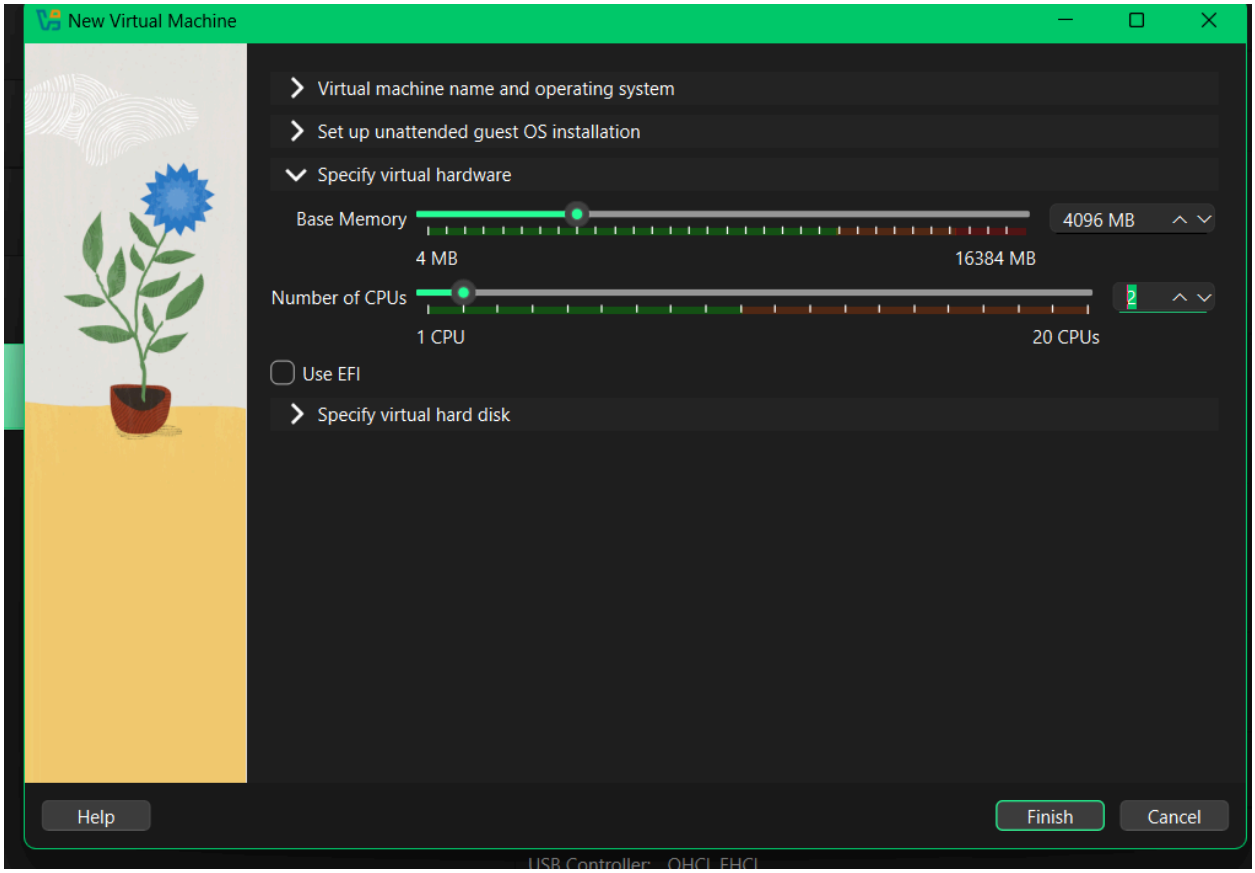
Screenshot 1: (Virtual Machine Setup Below).



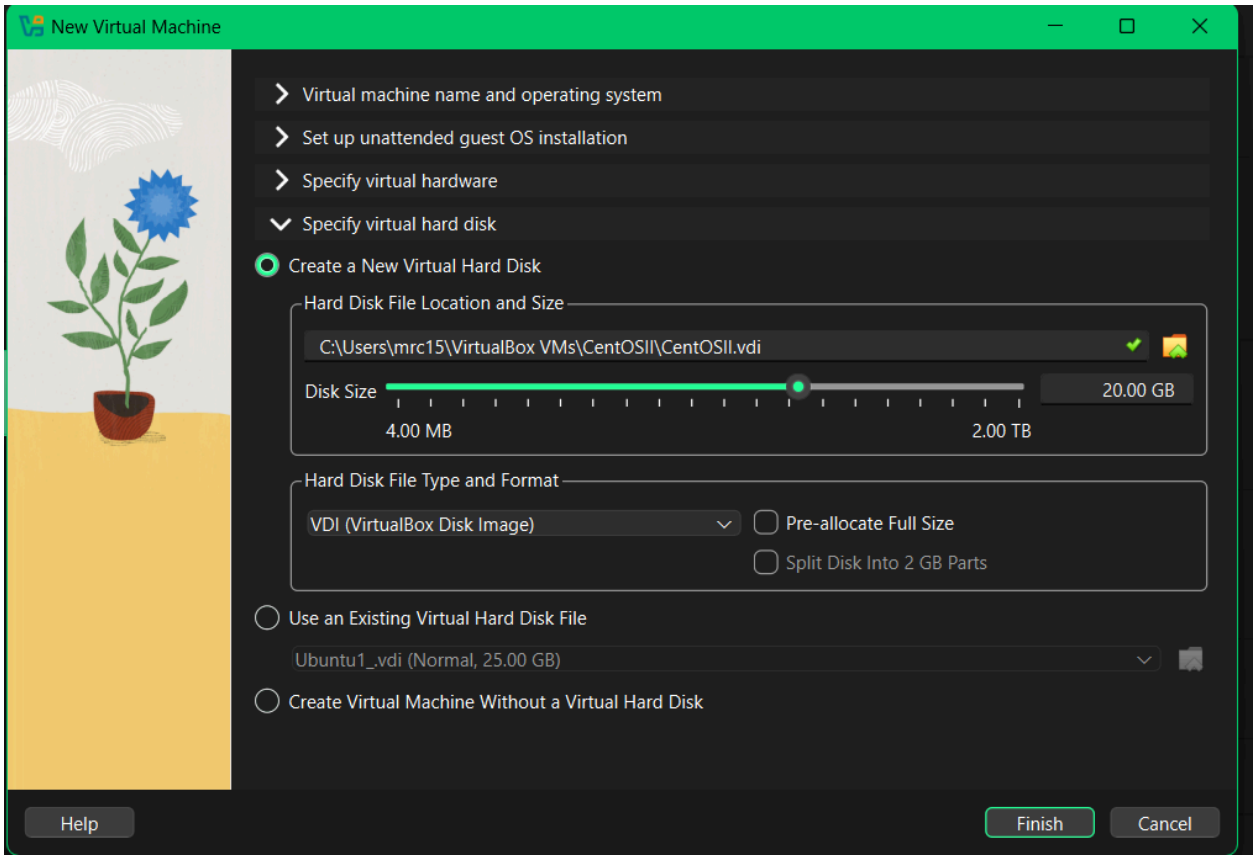
It is important to **name** your virtual machine, in this example the virtual machine is named **"CentOSII"**. Make sure to add the **CentOS iso, and get it from your download folder.**



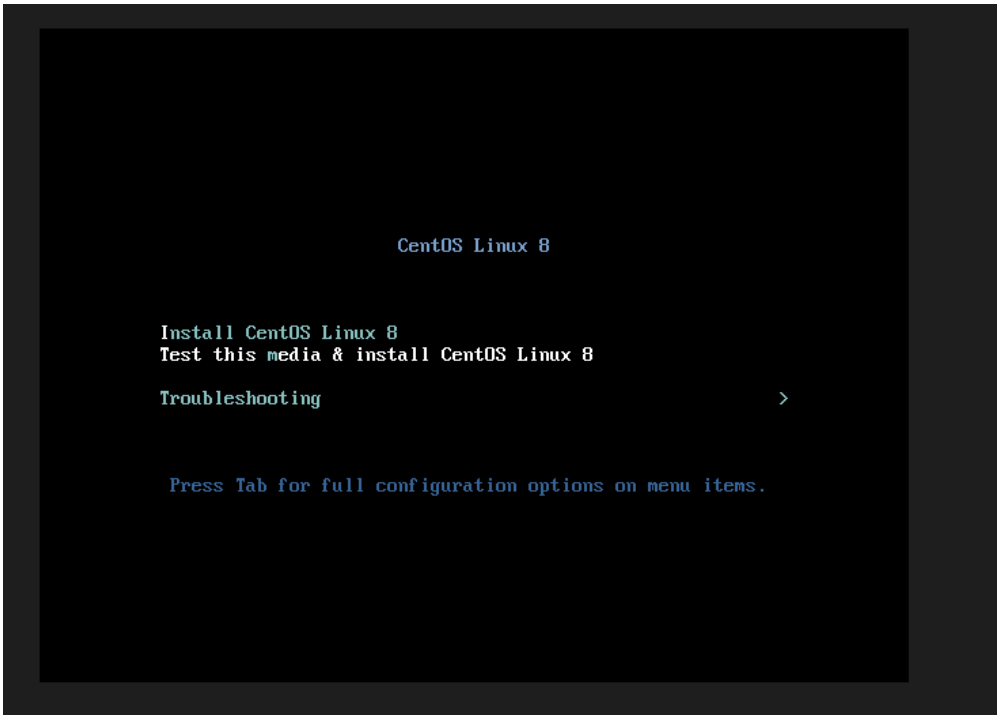
The proceeding screen is the password screen. On this screen you can see the **hostname** which is the name of your virtual machine. In this example the virtual machine name is **CentOSII,** under username it is set up as **vboxuser by default** you can enter your name like **Angel, Bobby, Jordan, etc. The password you enter can be a mix of lowercase, uppercase letters and numbers/symbols can be used.**



For the memory and cpu steps, it is **recommended** to have a **base memory size doubled** of that of the original, **which is 2048 *2 would be 4096 for base memory.** The original cpu count is **1,** it is a good idea to change the number. A **higher count of cpu can be better for speed and processing.** For this example the cpu count is **2.** It is important to have **4 processors for faster load times** for the server. **The default is two** and makes the server run slower for different downloads, it is recommended to use **double** the original intended amount.

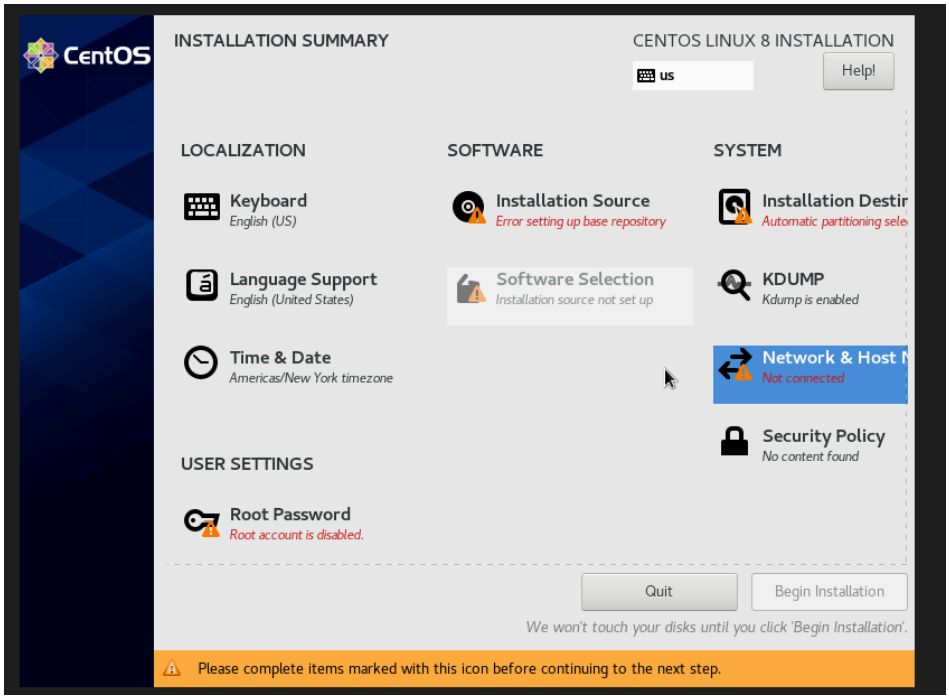


For the disc size it can stay as the default, **it can be changed to either 25, 30 or 35 GB if the initial 20 is not enough capacity.**

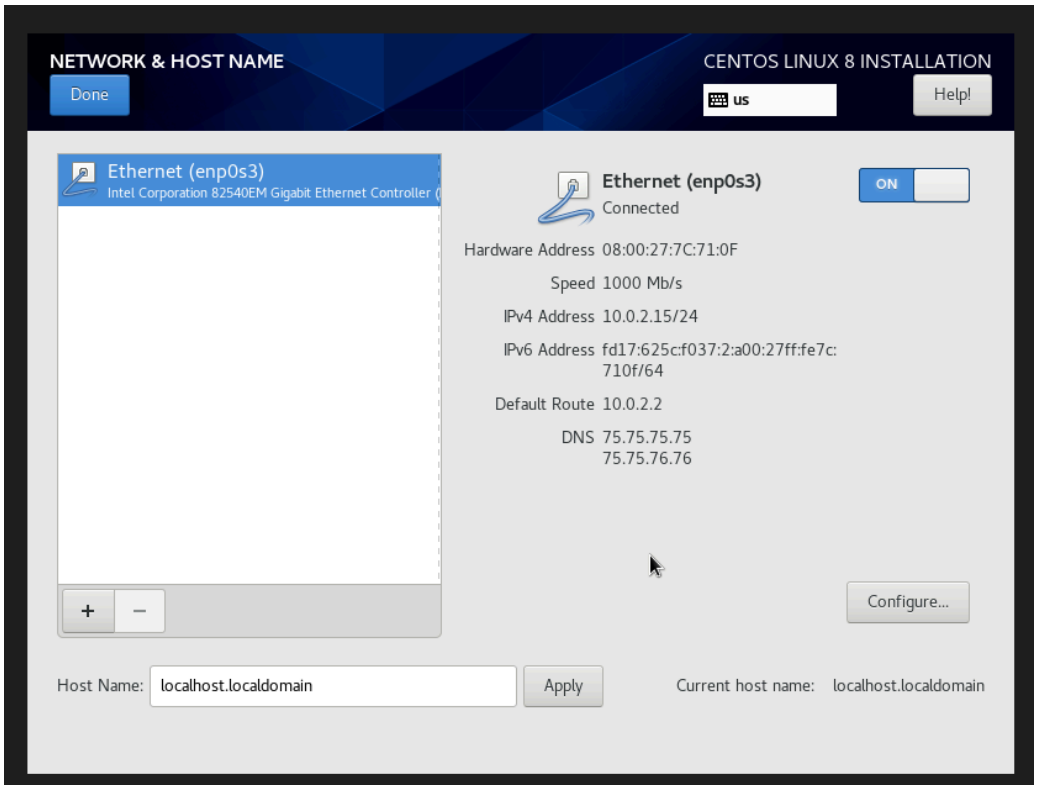


The very first screen you will see for the **CentOS server** is one saying **CentOS Linux 8.** There are a few options, the option you want to choose is **"Test this media & install CentOS Linux 8.**

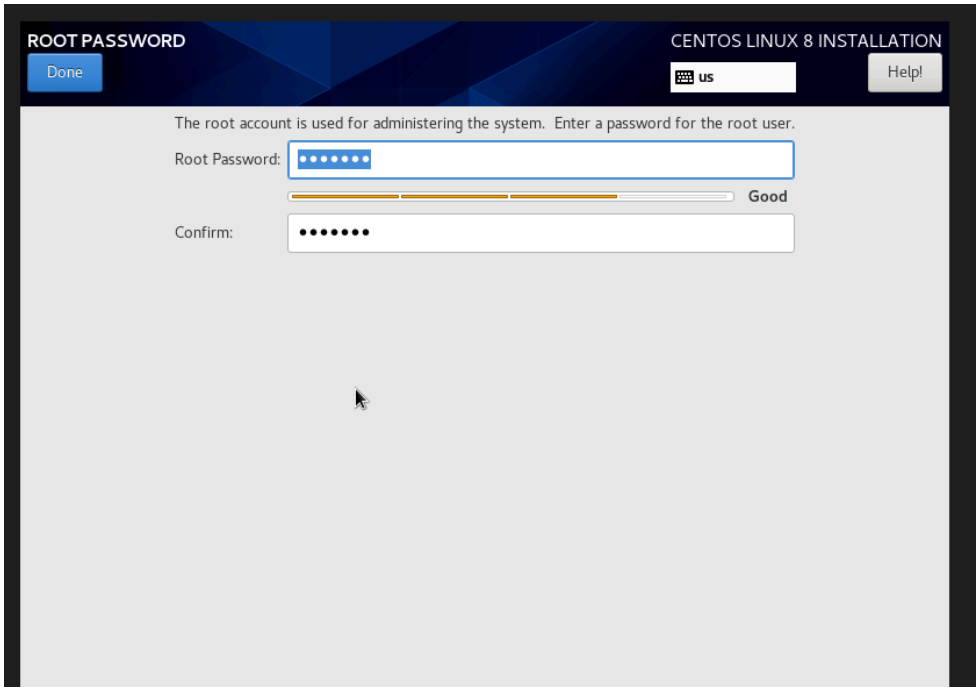
After pressing the test and install screen you will show up at this Welcome screen.**You want to use your preferred language for the server,** for this example I have chosen **English.**



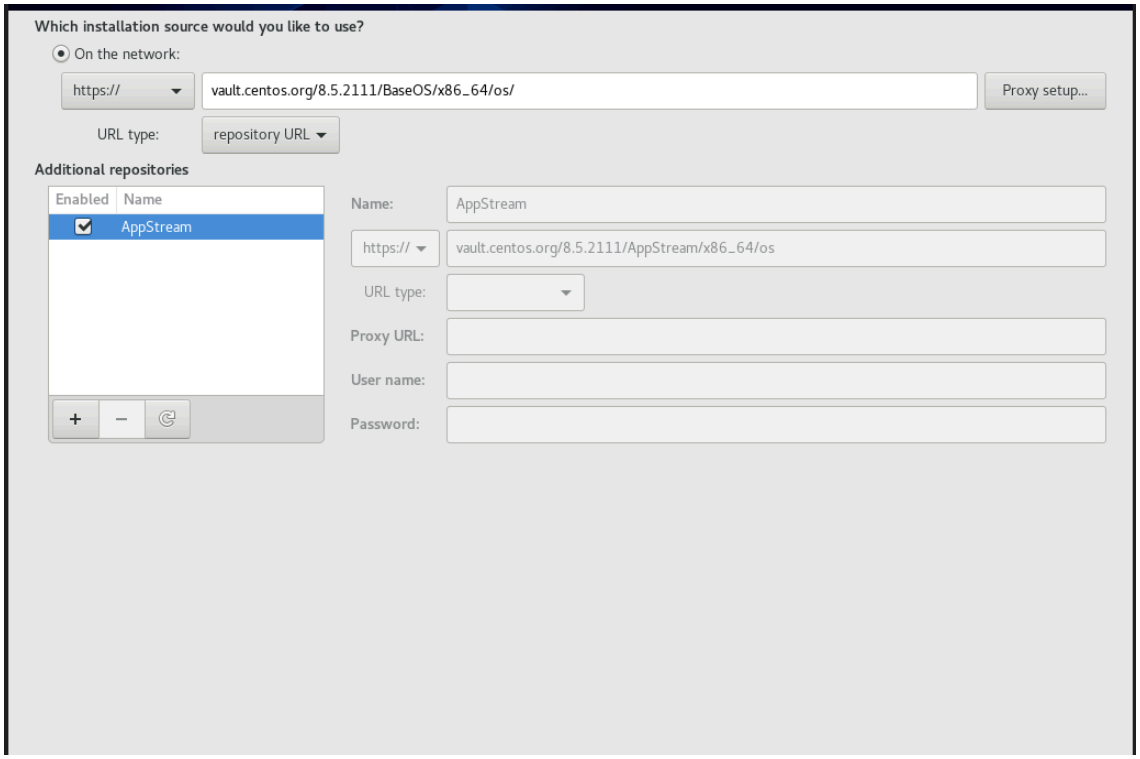
Upon choosing your preferred language you will be brought to the installation screen. This screen shows that there are **errors and steps that must be taken to proceed**. There are many options at this screen, you can choose any order. For this example the order I chose was **"Network & Host Name"**, then proceeding with the **"Root Password"**, "**Repository Source",** and finally **"Additional Software"**page.



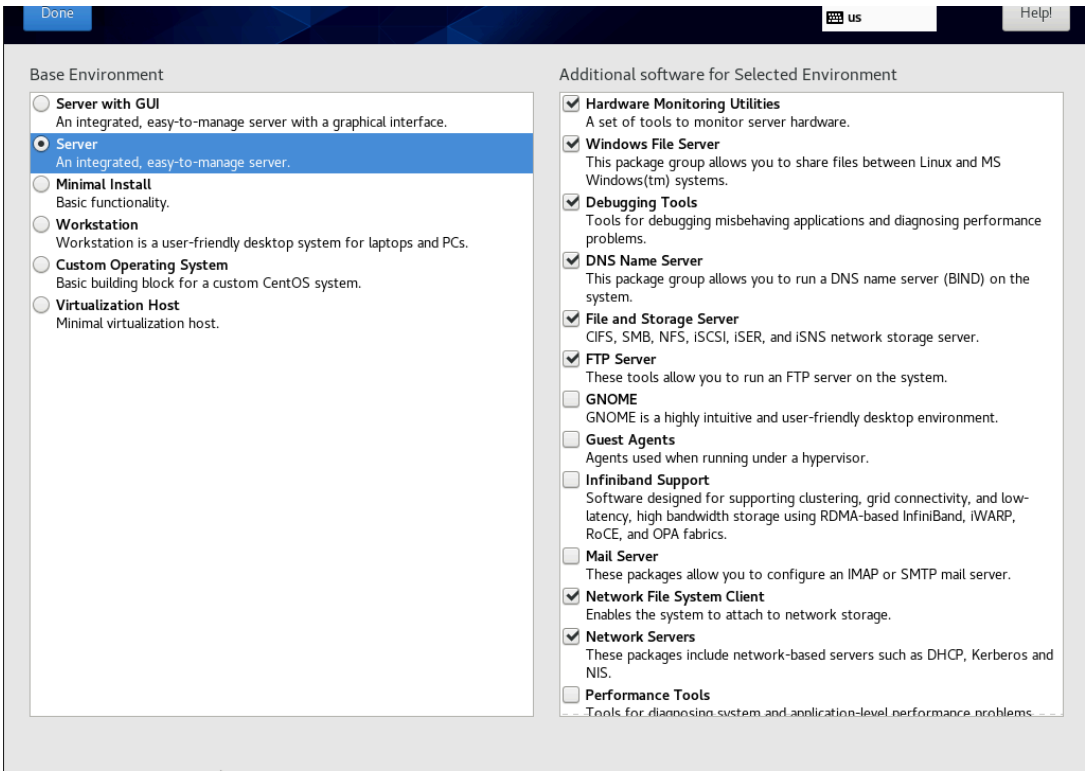
At the "**Network & Host Name"** screen there should be an **Ethernet cable.** By default this setting is disabled, you want to turn this setting on, **Ethernet on** to continue the installation of different services for the **CentOS Server.**



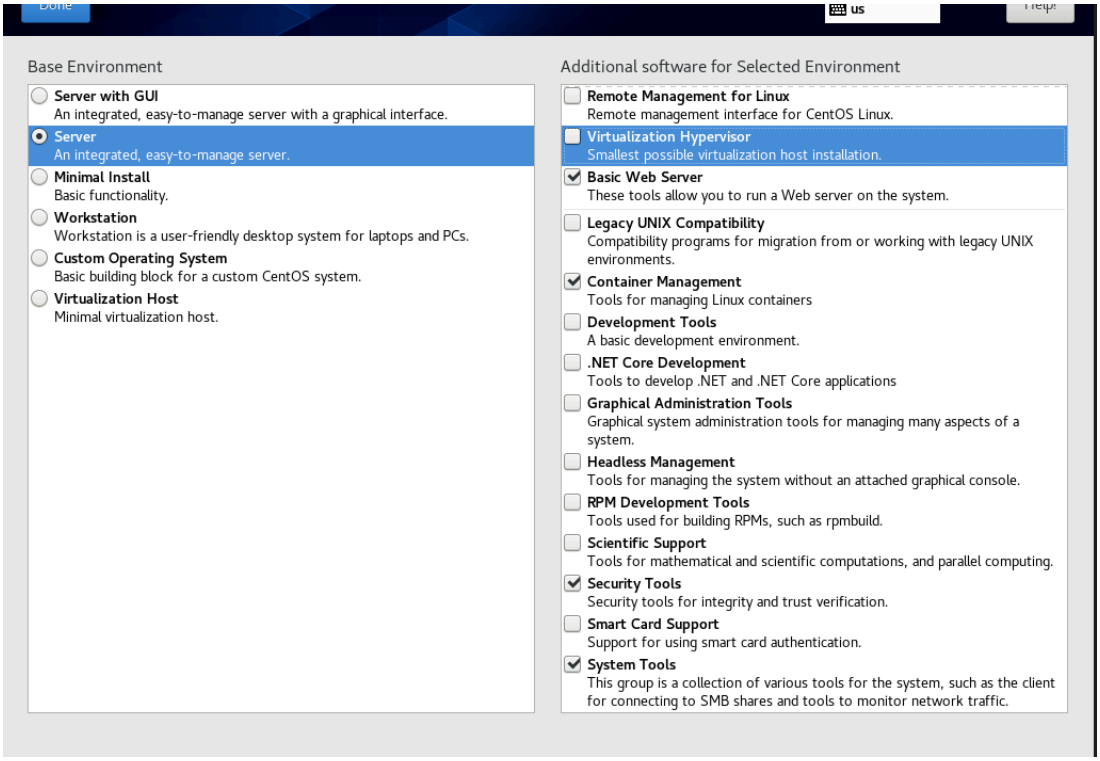
The next step is to set up your **Root Password.** This root password is **linked to the user** you created when creating the **CentOS** server within a virtual machine environment like **virtualbox.**



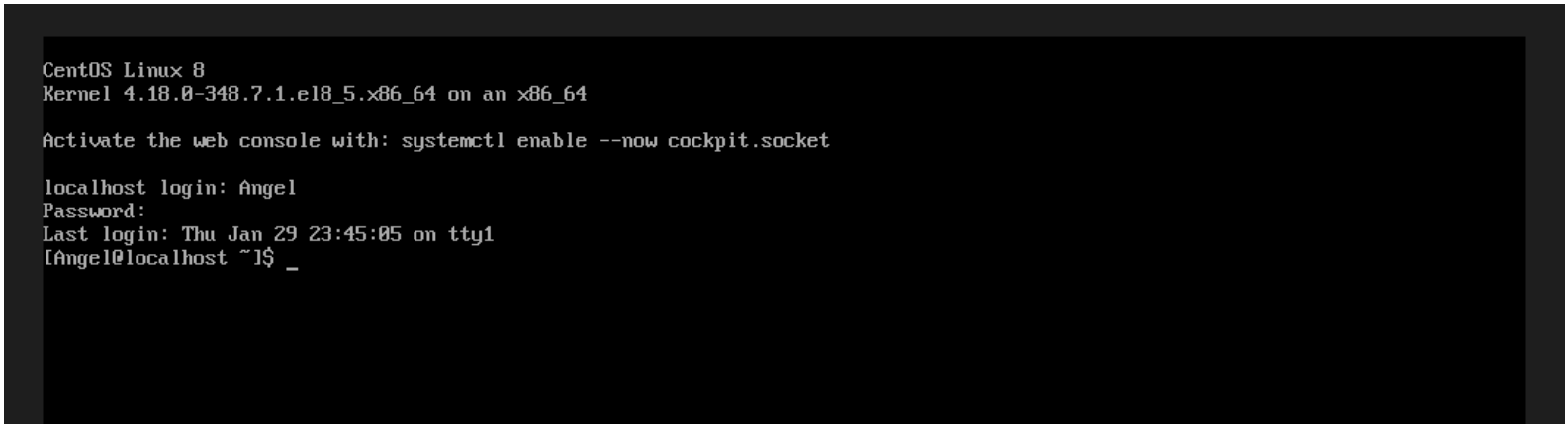
The following screenshot above for **CentOS** shows the system is connected to the network, the url type and what repository the virtual machine is connected to. **This must be written to continue the setup.**



This screenshot shows the additional software that would be used for the first distro **"CentOS". It is important to use the server as the environment, as using the server you can control different aspects of the virtual machine.** The screenshot above shows the additional software that must be used for the server to run properly, they are check marked boxes in order.



The screenshot above shows a continuation of the additional software that was used for the first distro **CentOS. These continued check marked boxes are the final additions that must be added to your server, to run properly.**

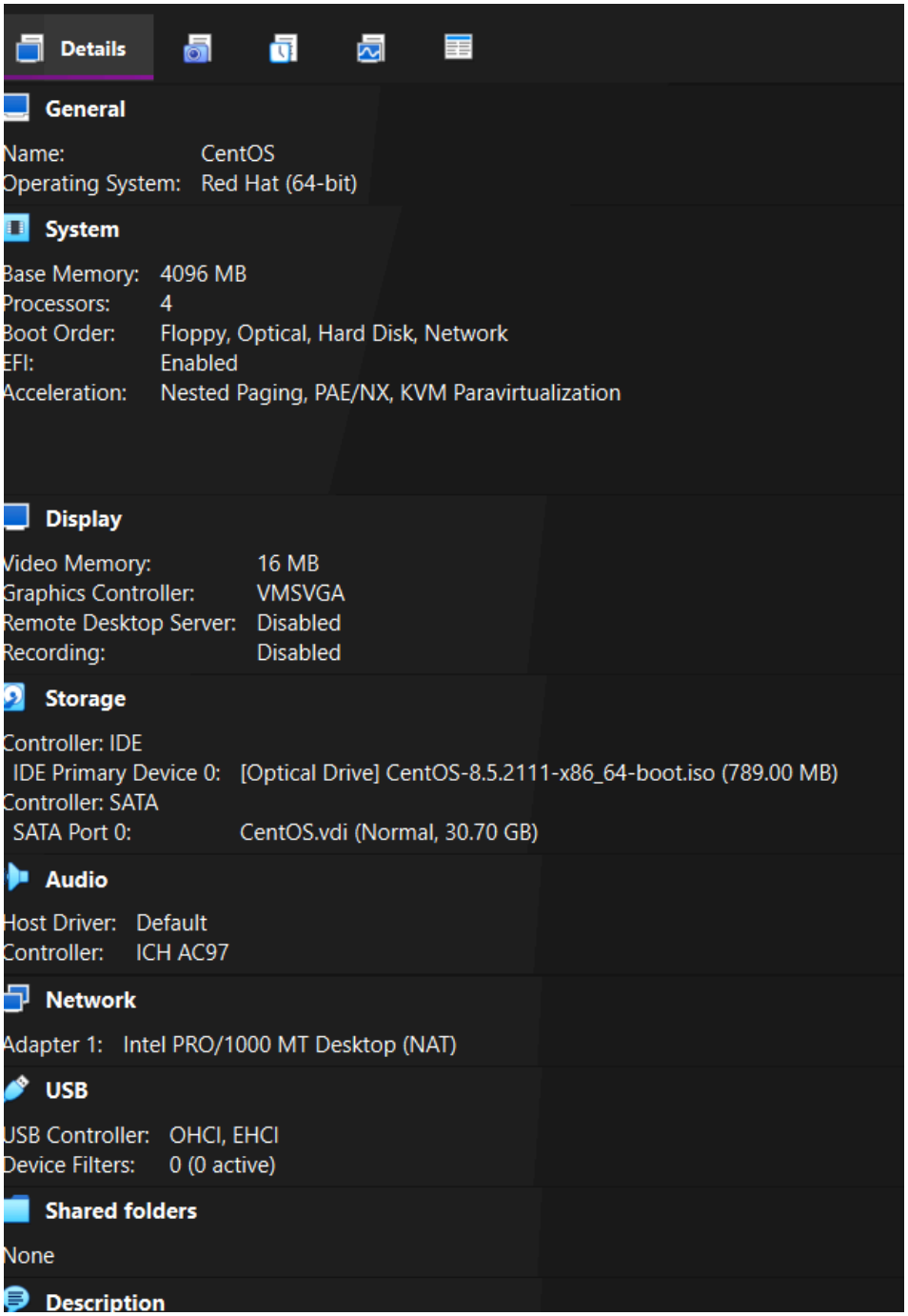


```
[root@localhost ~]# useradd Angel
[root@localhost ~]# passwd Angel
Changing password for user Angel.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
[root@localhost ~]# passwd Angel
Changing password for user Angel.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# usermod-aG wheel username Angel
-bash: usermod-aG: command not found
[root@localhost ~]# usermod -aG wheel username Angel
Usage: usermod [options] LOGIN

Options:
  -c, --comment COMMENT         new value of the GECOS field
  -d, --home HOME_DIR           new home directory for the user account
  -e, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -f, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -g, --gid GROUP               force use GROUP as new primary group
  -G, --groups GROUPS           new list of supplementary GROUPS
  -a, --append                  append the user to the supplemental GROUPS
                                mentioned by the -G option without removing
                                the user from other groups
  -h, --help                    display this help message and exit
  -l, --login NEW_LOGIN         new value of the login name
  -L, --lock                    lock the user account
  -m, --move-home               move contents of the home directory to the
                                new location (use only with -d)
  -o, --non-unique              allow using duplicate (non-unique) UID
  -p, --password PASSWORD       use encrypted password for the new password
  -R, --root CHROOT_DIR         directory to chroot into
  -P, --prefix PREFIX_DIR       prefix directory where are located the /etc/* files
  -s, --shell SHELL             new login shell for the user account
  -u, --uid UID                 new UID for the user account
  -U, --unlock                  unlock the user account
  -v, --add-subuids FIRST-LAST  add range of subordinate uids
  -V, --del-subuids FIRST-LAST  remove range of subordinate uids
  -w, --add-subgids FIRST-LAST  add range of subordinate gids
  -W, --del-subgids FIRST-LAST  remove range of subordinate gids
  -Z, --selinux-user SEUSER     new SELinux user mapping for the user account

[root@localhost ~]#
```

Once the server is installed, to make a new user you would write this command, **ls useradd (yourname)**. To create a password for the user you would write this command, **passwd for (yourusername).** The system will then ask you to write your **unique password,** which must be **8 characters in length.** If the password is less than **8 characters the system will give a bad password error.**

Final Screenshot: (System Requirements for CentOS).

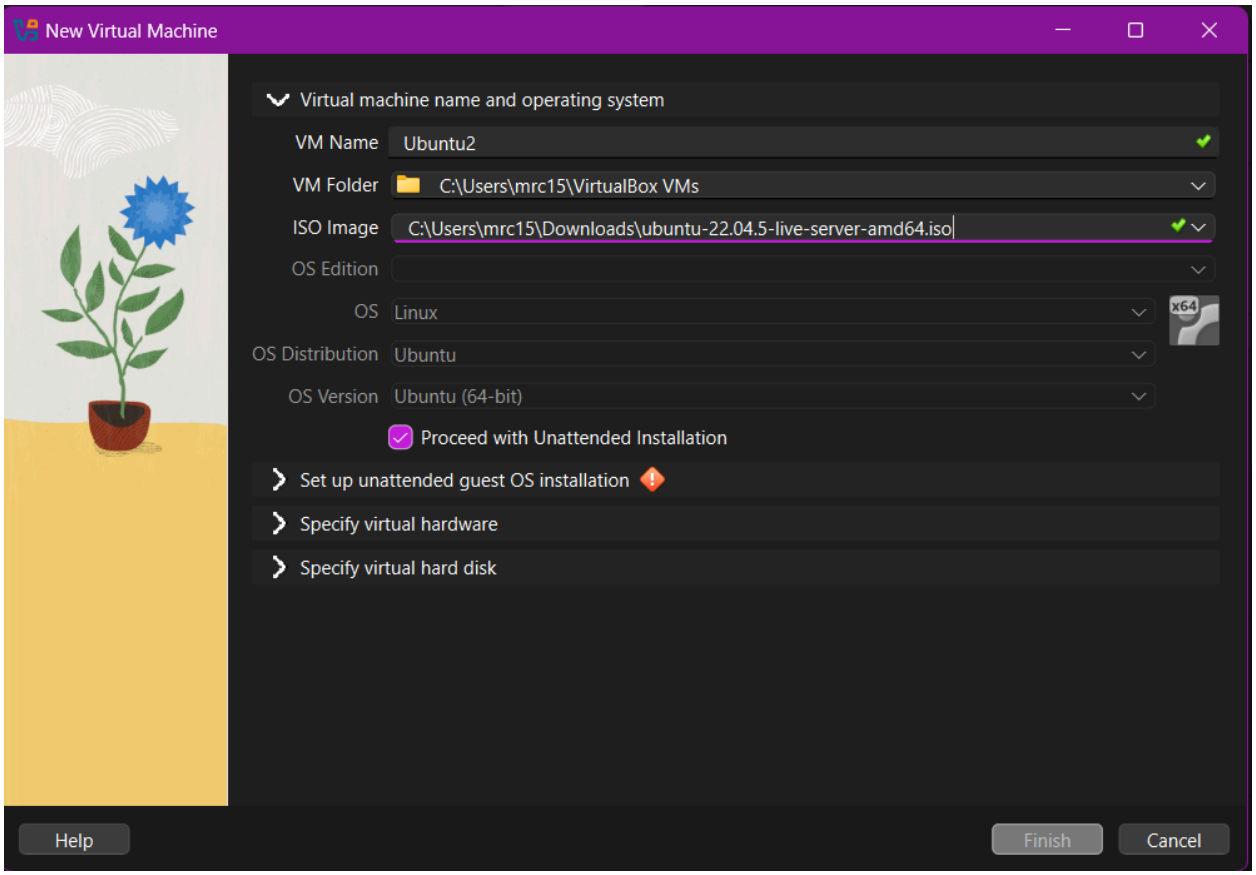
(Below).



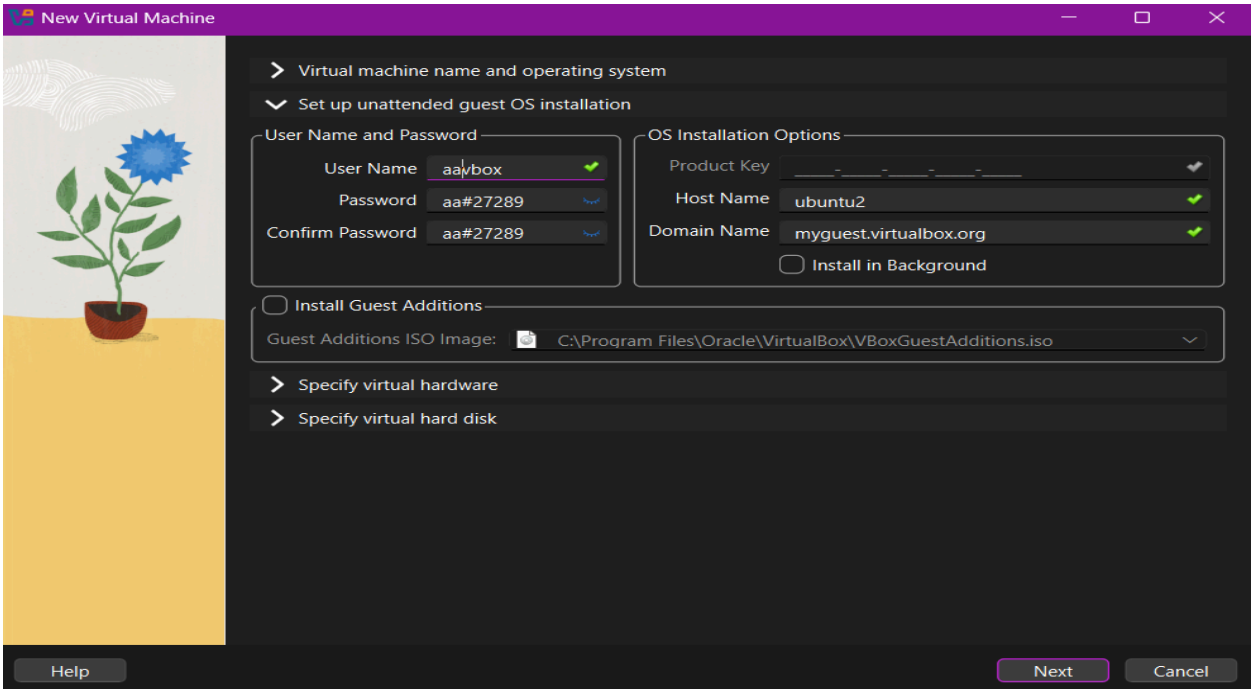
Ubuntu Server Source- https://releases.ubuntu.com/jammy/

For this source I could not find one that worked properly within the virtualbox environment. I downloaded different versions of the iso and eventually got one to work. The error I was receiving was not a **large enough disc size or a disc being nonexistent**.This process took me a long time to figure out and have the correct settings. **I first downloaded Debian and it did not work and then I downloaded Ubuntu. Ubuntu worked properly.**

Screenshot 1:

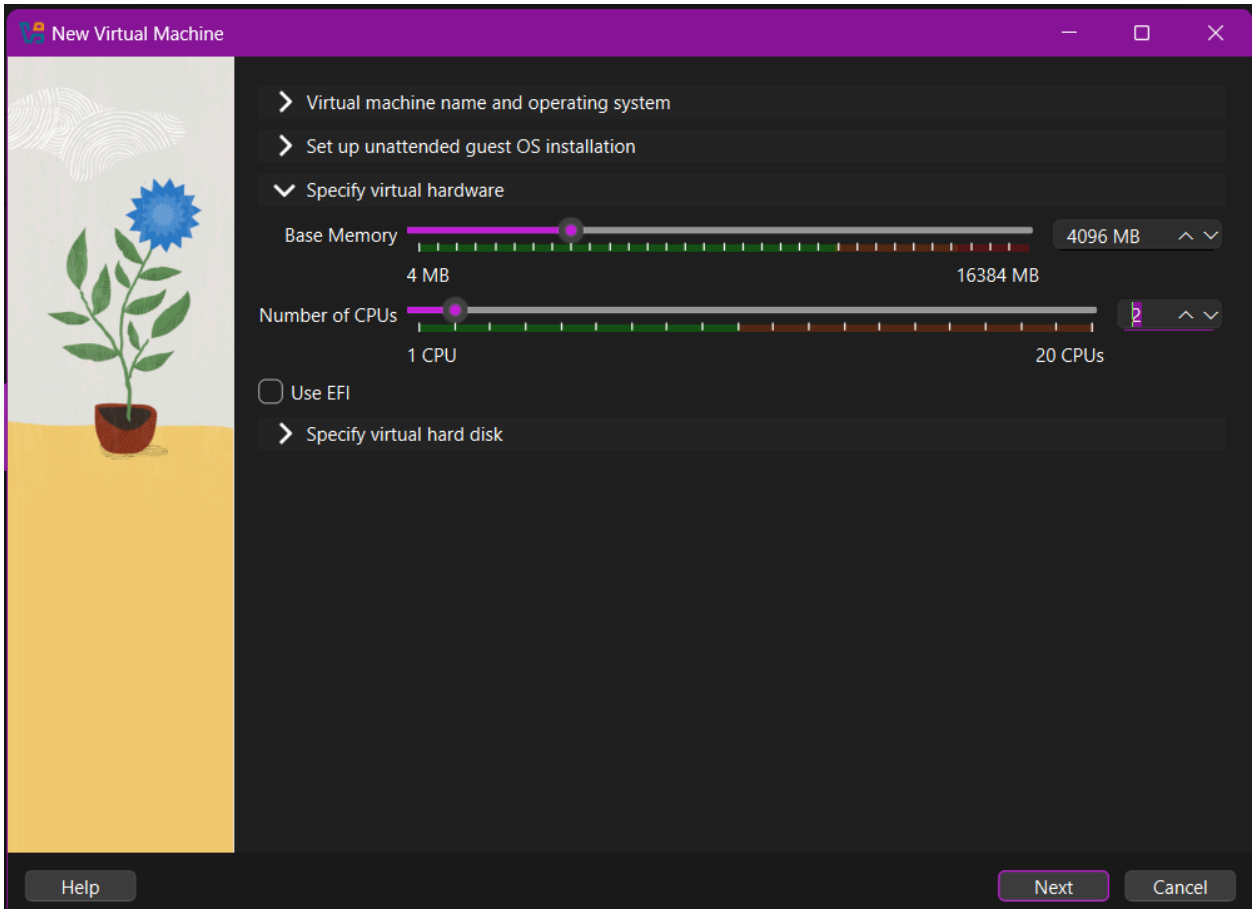


For the initial setup of the **Ubuntu Server**, a user should first name the server, and make sure where the download source folder is for their iso**. For the virtual machine the name cannot contain any spaces or symbols, you can only add a(dash -) or numbers to the end** of

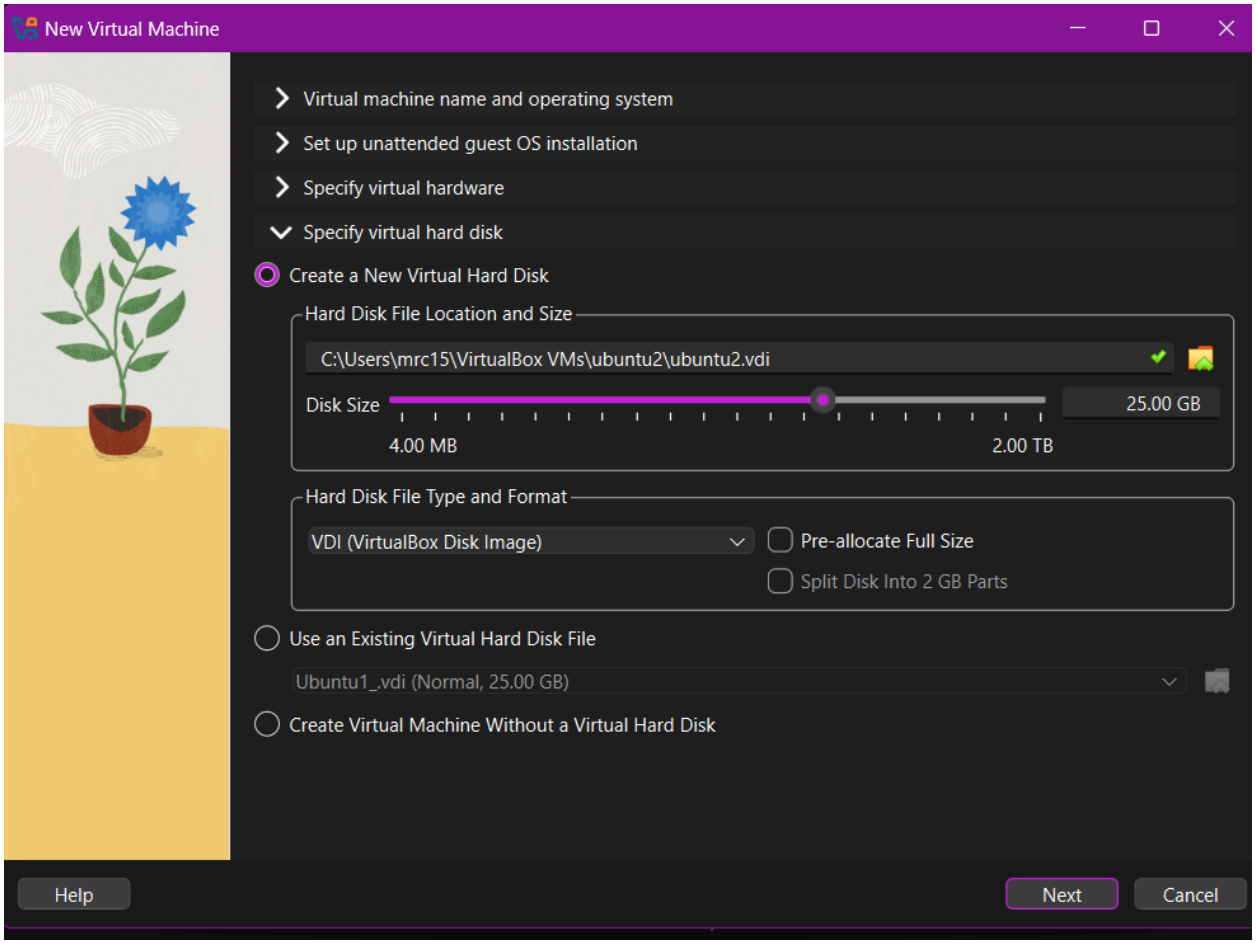


the name created. **From download you would then click iso and click on the iso you are using like Debian, Ubuntu, CentOS etc.**

For the next step the user would then create a unique username and password, confirm the domain name is accurate and the host name is set to your preferred name. For this example, **my username is aavbox, my password is aa#27289.** The setup requires that you have a username and password with a minimum length of **8 characters.**

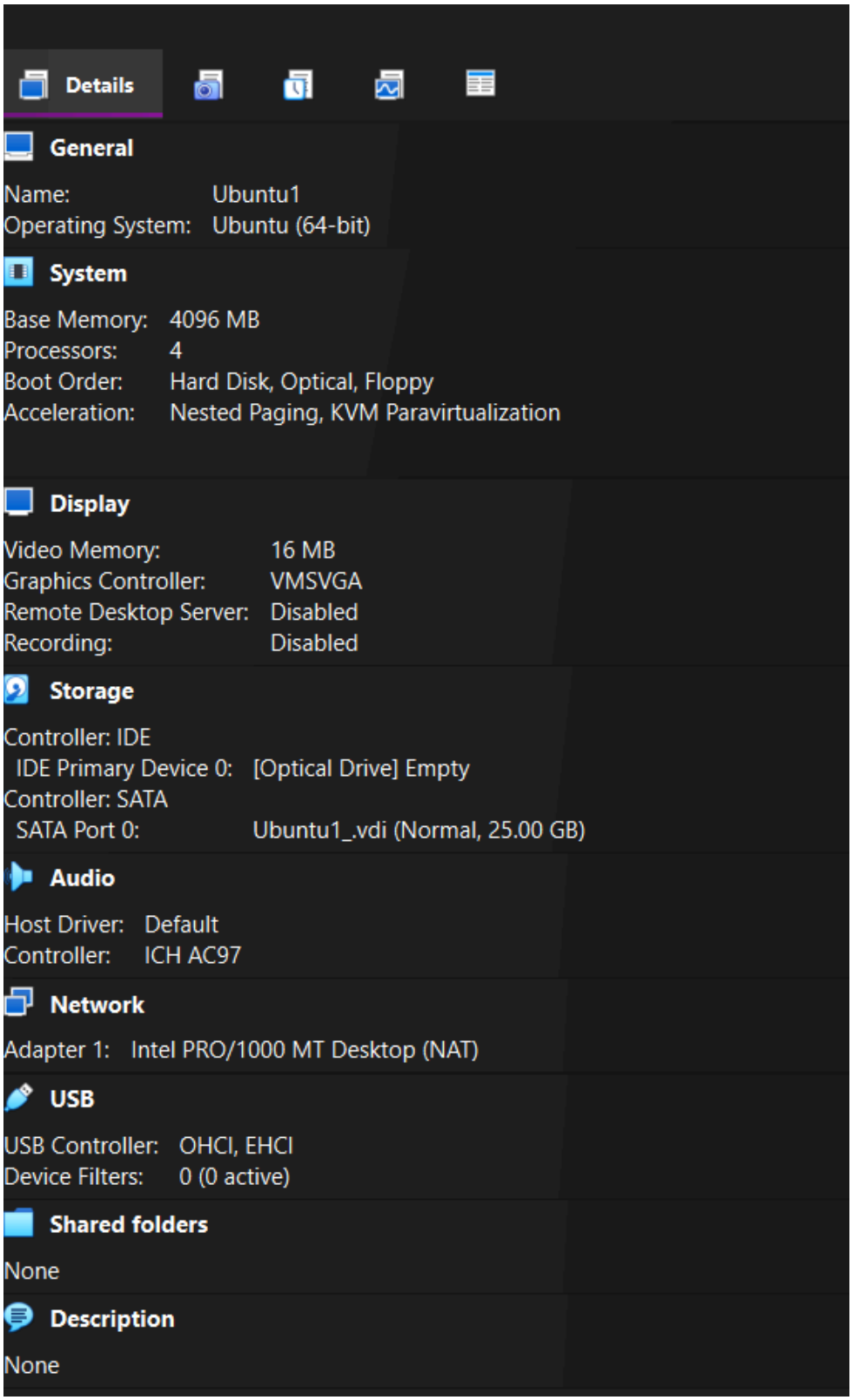


For this step it is **recommended** to have a **base memory size doubled** of that of the original, **which is 2048 *2 would be 4096 for base memory.** The original cpu count is **1,** it is a good idea to change the number. With a **higher cpu count you will get better speeds and processing.** For this example the cpu count is **2.** It is important to have **4 processors for faster load times** for the server. **The default is two** and makes the server run slower for downloads, so it is recommended using **double** the original amount of processing power.



For the disc size it can stay as the default, **it can be changed to either 30, 35 or 40 GB if the initial 25 is not enough capacity.**

System Requirements Screenshot: (Ubuntu Server).



```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 28 07:46:48 PM UTC 2026

  System load:             0.34
  Usage of /:              13.2% of 24.44GB
  Memory usage:            5%
  Swap usage:              0%
  Processes:               113
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe5c:2b44

Expanded Security Maintenance for Applications is not enabled.

69 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

aavbox26@Ubuntu1:~$
aavbox26@Ubuntu1:~$
```

Once the server is installed the program will ask you to **sign into the system with your unique specified user. The system will also ask for the password,** afterwards you can start writing commands in the Linux server.

Question **3:**

I used the server options for the setup of both distros. I made sure I chose the correct source. I was using the server and not the desktop operating systems version of the two linux distros, **CentOS** and **Ubuntu.**

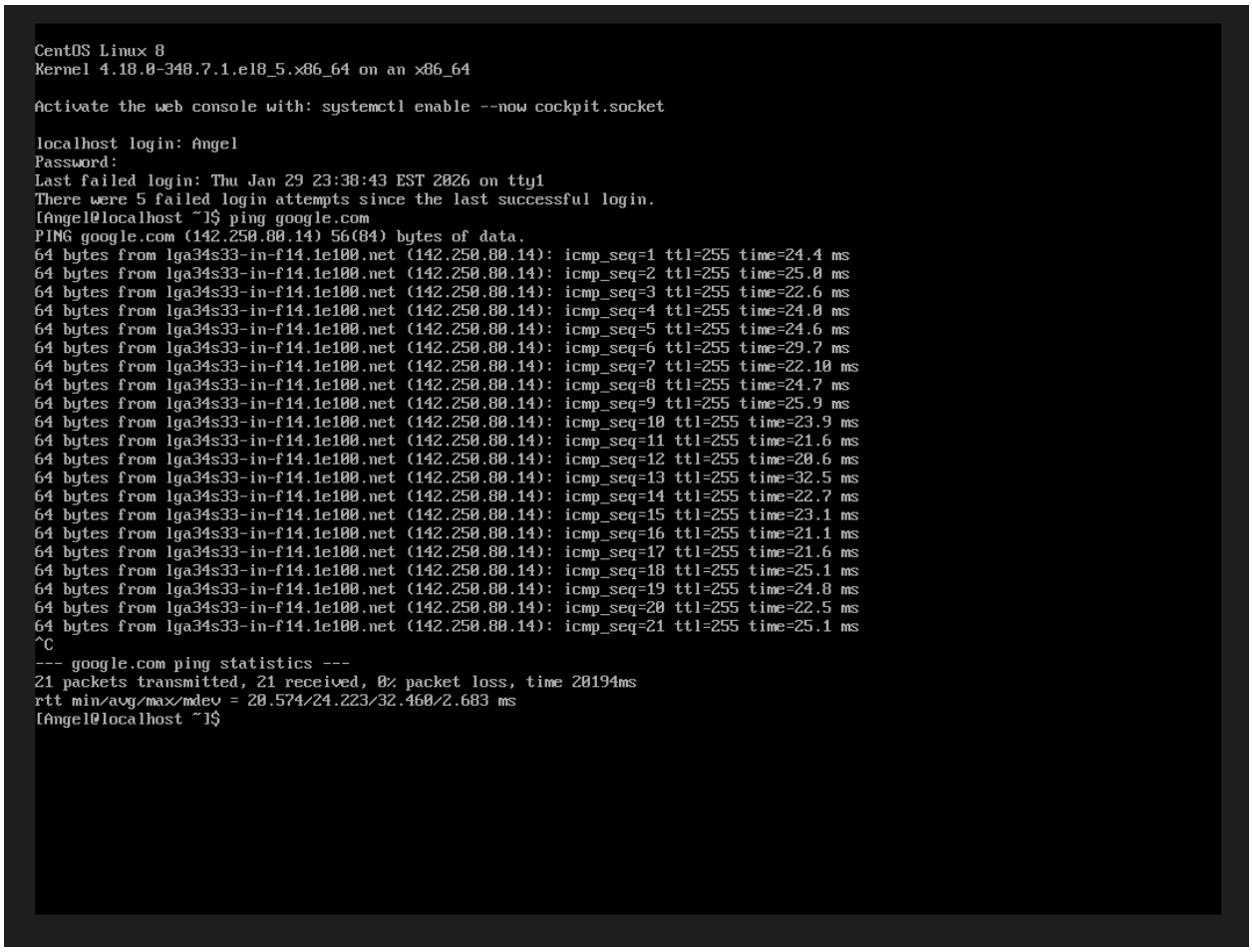
Question **4:**

I installed everything without using a GUI or minimal install. **The linux servers use command prompts as intended.**

Question **5:**

I had made sure that both servers had internet access and tested them. Below I have added screenshots of the test for both linux virtual machines. I pinged both **CentOS** and **Ubuntu to ping** google.com.

Screenshot 1: CentOS Ping.



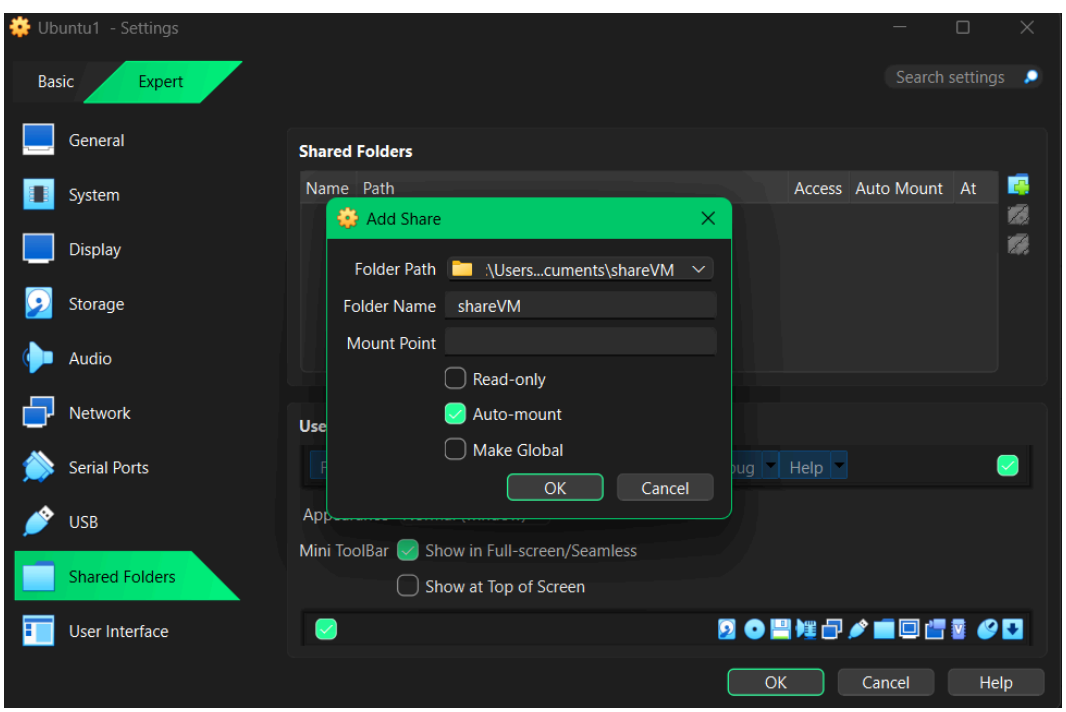
```
CentOS Linux 8
Kernel 4.18.0-348.7.1.el8_5.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

localhost login: Angel
Password:
Last failed login: Thu Jan 29 23:38:43 EST 2026 on tty1
There were 5 failed login attempts since the last successful login.
[Angel@localhost ~]$ ping google.com
PING google.com (142.250.80.14) 56(84) bytes of data.
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=1 ttl=255 time=24.4 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=2 ttl=255 time=25.0 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=3 ttl=255 time=22.6 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=4 ttl=255 time=24.0 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=5 ttl=255 time=24.6 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=6 ttl=255 time=29.7 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=7 ttl=255 time=22.10 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=8 ttl=255 time=24.7 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=9 ttl=255 time=25.9 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=10 ttl=255 time=23.9 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=11 ttl=255 time=21.6 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=12 ttl=255 time=20.6 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=13 ttl=255 time=32.5 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=14 ttl=255 time=22.7 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=15 ttl=255 time=23.1 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=16 ttl=255 time=21.1 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=17 ttl=255 time=21.6 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=18 ttl=255 time=25.1 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=19 ttl=255 time=24.8 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=20 ttl=255 time=22.5 ms
64 bytes from lga34s33-in-f14.1e100.net (142.250.80.14): icmp_seq=21 ttl=255 time=25.1 ms
^C
--- google.com ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 20194ms
rtt min/avg/max/mdev = 20.574/24.223/32.460/2.683 ms
[Angel@localhost ~]$
```

Screenshot 2: Ubuntu Ping.

```
PING google.com (142.251.41.174) 56(84) bytes of data.
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=1 ttl=255 time=25.3 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=2 ttl=255 time=27.1 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=3 ttl=255 time=26.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=4 ttl=255 time=26.5 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=5 ttl=255 time=23.4 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=6 ttl=255 time=27.0 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=7 ttl=255 time=23.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=8 ttl=255 time=24.9 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=9 ttl=255 time=28.6 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=10 ttl=255 time=26.5 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=11 ttl=255 time=21.8 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=12 ttl=255 time=24.4 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=13 ttl=255 time=25.0 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=14 ttl=255 time=27.6 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=15 ttl=255 time=27.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=16 ttl=255 time=26.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=17 ttl=255 time=26.3 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=18 ttl=255 time=24.9 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=19 ttl=255 time=25.1 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=20 ttl=255 time=22.4 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=21 ttl=255 time=26.3 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=22 ttl=255 time=23.9 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=23 ttl=255 time=23.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=24 ttl=255 time=26.8 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=25 ttl=255 time=26.6 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=26 ttl=255 time=27.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=27 ttl=255 time=24.7 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=28 ttl=255 time=25.6 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=29 ttl=255 time=24.2 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=30 ttl=255 time=30.3 ms
64 bytes from lclgaa-at-in-f14.1e100.net (142.251.41.174): icmp_seq=31 ttl=255 time=21.9 ms
^C
--- google.com ping statistics ---
31 packets transmitted, 31 received, 0% packet loss, time 31509ms
rtt min/avg/max/mdev = 21.840/25.486/30.333/1.907 ms
aavbox26@Ubuntu1:~$
```

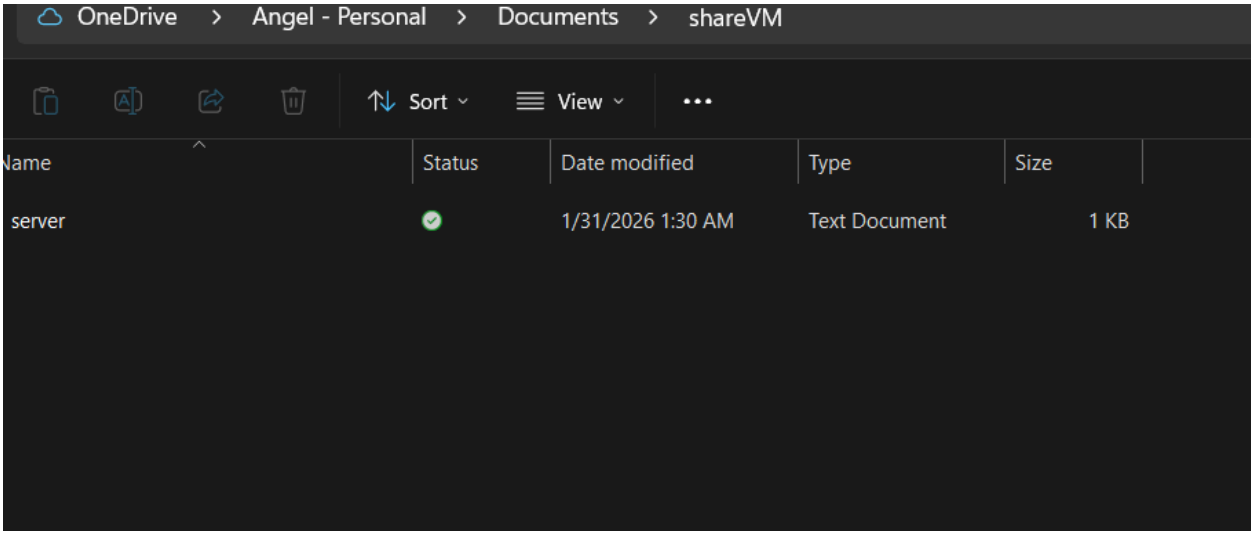
Final Question; Question **#6:**

Below I have added screenshots of how to use a file share method, and how I completed the setup with step by step instructions.

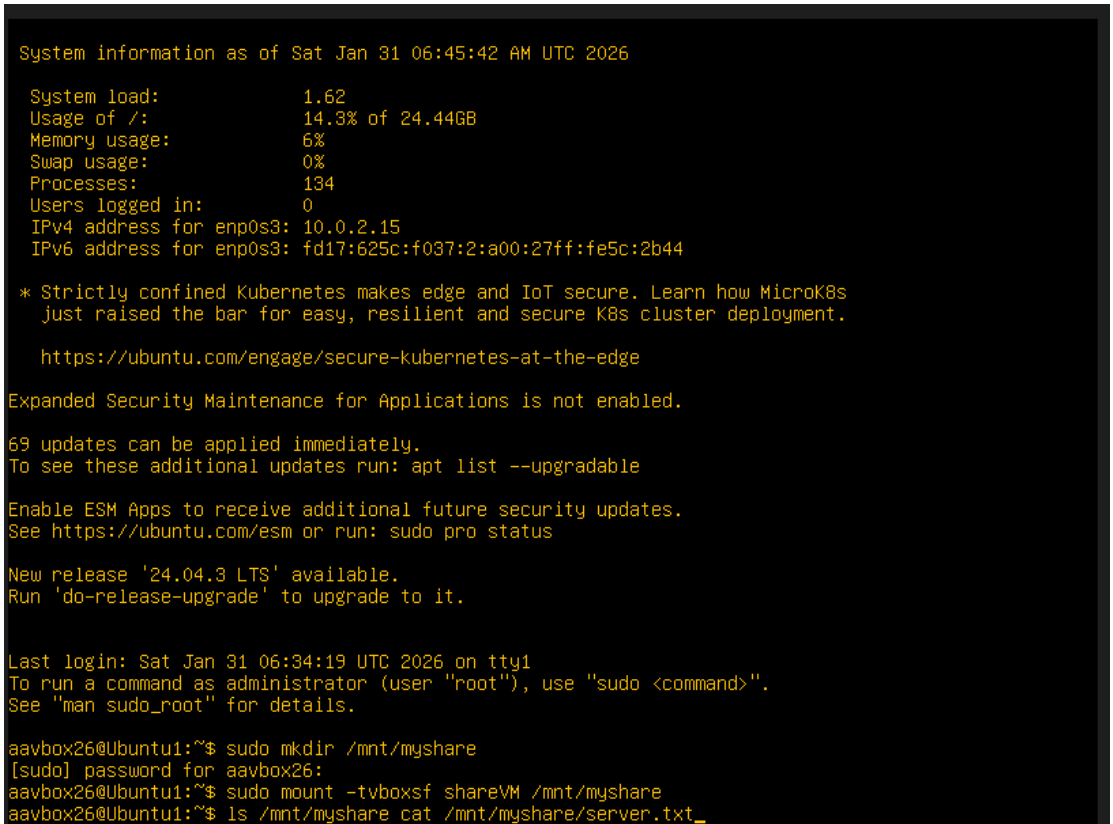


The first thing I did for the file share was, create a folder on my windows PC. I named the folder **shareVM,** short for **"Share to Virtual Machine Folder".** Then I **auto-mounted** the folder so it can be **read in the Ubuntu and CentOS Server.** You have to do this separately for both virtual servers.

When you check off **Auto-mount,** the system creates a permanent connection between the host computer and a directory that is **located in media/sf_shareVM.** This makes the file appear each time without mounting it. **This only worked with Ubuntu for some reason it did not work within the CentOS environment, CentOS** blocked this action.



The text file within the folder named share VM was named server.txt. When writing the word and then **txt** the file just names it server for short and converts it to a text file for systems to read.



In the image above I use three different commands. The first command is **sudo mkdir /mnt/myshare.** After writing this command the system will then ask for your **password** for the **user you created within the Ubuntu server** environment. For this example, the username is **aavbox26, the password contains 8 characters with a mix of letters, numbers, and symbols but cannot be seen as Ubuntu does not show anything for passwords.**

The second command written in this image was **sudo mount -t vboxsf shareVM /mnt/myshare.** **Example for commands:** **https://help.ubuntu.com/community/VirtualBox/SharedFolders**

The final commands written in this image for file sharing was **ls /mnt/myshare,** then proceeding it in the same line was **cat /mnt/myshare/server.txt.** **These three commands link the text file to the Ubuntu server.**

Example for commands: **https://www.virtualbox.org/manual/ch04.html#sf_mount_auto**

For the Ubuntu server, to see the text file that was written in the host computer environment was this command **ls /mnt/myshare.** **The server then shows that the text file is within the Ubuntu environment.**

Example for command:
https://www.virtualbox.org/manual/ch04.html#sf_mount_auto

The following command shows the text written within the text file, within the Ubuntu server environment which is **cat /mnt/myshare/server.txt.** The Ubuntu server then reads the text, which is **"Hello Ubuntu, CentOS Server!".**

Important Things to remember, Linux:

Ls- Lists. Shows all files etc.

mnt - Mount. Links to external drives or shared folders.

Myshare- Mount point, a folder created to share files.

Cat- Concatenate. A command that opens files and displays them, like "Hello Server!".

.txt- Creating Text files.

Sudo- Runs things with administrative controls and settings.

Mkdir- Make a directory, a command to create folders.

-t - Means type. Tell Linux machines what type of file you are connecting to.

Vboxsf- An argument telling something what to do. Mounts shared folders.
https://docs.oracle.com/en/virtualization/virtualbox/6.0/user/sharedfolders.html

Media- A directory for devices, used as a mount point for storage devices.

Root- High level of authority in virtual machines. Name for an account with the highest access like a username that is an administrator for a server or virtual server environment.

The method I chose for the **CentOS** server shown in the screenshot below was, writing the file manually with **nano** within the **CentOS** server. The file was able to be read but it is not the original document that was also connected to **CentOS.** The system within **CentOS** continued to give errors like password invalid when trying to connect to **Ubuntu** or the file did not exist even though the file was also Auto-mounted to **CentOS.**